

● HARD

● INFORMATION AND IDEAS

● ANSWER KEY

Information and Ideas

12

10 answers · Rationale-rich · Teach from doc

Q1**A**

- A. It is less suitable for manufacturing large volumes of planar p-i-n perovskite solar cells than an alternative fabrication method may be.
-

Choice A is the best answer. Conventional solar cell fabrication increases “the cost of manufacturing at scale,” but spray coating might get rid of that problem.

Choice B is incorrect. This is not completely supported by the text. While it’s true that conventional ETL fabrication is expensive at scale, there’s nothing in the text that mentions other perovskite solar cell architectures. Choice C is incorrect. This choice does not match the text. Only one conventional method of ETL fabrication is described, so we can’t compare the solution loss in this method to that of other conventional methods. Choice D is incorrect. This choice isn’t supported by the text. The text never suggests that the effectiveness of solar cells changes based on their method of fabrication.

Q2

C

- C. says to himself while striking his head, “Beat at this gate that let thy folly in / And thy dear judgement out!”
-

Choice C is the best answer because it most effectively uses a quotation from *King Lear* to illustrate the claim that King Lear expresses regret for his actions. In the quotation, Lear describes striking himself on the head—the same act he’s engaged in as he speaks, and one that suggests he’s deeply upset with himself. Referring to himself in the second person (with “thy”), the character exclaims “Beat at this gate that let thy folly in / And thy dear judgement out!” Lear refers metaphorically to his own mind as a gate that has allowed folly, or poor judgement, to enter and good judgement to escape. This suggests that Lear regrets his attempts to test his three daughters’ devotion to him, regarding those attempts as examples of the folly that has entered the gate of his mind.

Choice A is incorrect because this quotation doesn’t express King Lear’s sense of regret over his own actions; instead, it expresses his belief that the harm that others have done to him (or the extent to which they have “sinned against” him) outweighs whatever harm he himself has caused by “sinning.” Choice B is incorrect because this quotation doesn’t express King Lear’s sense of regret over his own actions; instead, it expresses his thoughts about an approaching storm (“this tempest”), which he believes “will not give [him] leave to ponder,” or time to consider, the harm that he will continue to experience (“things” that “would hurt [him] more”). Choice D is incorrect because this quotation expresses King Lear’s vow to commit terrible actions (or “things” that “shall be / The terrors of the earth”) in the future, not his regret over actions that he’s already taken.

Q3

D

- D. the aerogel will be less prone to the structural weakness that ultimately causes most other aerogels to break down with use.
-

Choice D is the best answer because it most logically completes the text's discussion of aerogels. The text states that aerogels—highly porous foams—offer "excellent insulation" but typically break down after prolonged exposure to high heat. However, according to the text, Duan and colleagues developed an aerogel that "contracts rather than expands when heated" and recovers its original volume after this contraction. Thus, it is logical to conclude that Duan's team's aerogel material will be less prone to the structural weakness that caused earlier aerogels to break down.

Choice A is incorrect. Although the text indicates that aerogels consist "mainly of tiny air pockets within a solidified gel," it doesn't mention the number or proportion of air pockets to solidified gel in typical aerogels or in the aerogel developed by Duan's team. Choice B is incorrect because the text suggests that the aerogel developed by Duan's team has a higher, not a lower, ability to withstand exposure to intense heat due to its contraction and subsequent recovery. Choice C is incorrect. Although the text discusses temperature tolerances of aerogels and says that they offer "excellent insulation despite typically being brittle and eventually fracturing," it doesn't discuss how different rates of temperature change can affect aerogels.

Q4

C

- C. Research suggests that users with high expectations for a new technology can feel content with that technology even after experiencing negative disconfirmation.
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Choice C is the best answer. The passage begins by describing the study, and concludes with its results: "many users employed cognitive mechanisms to mitigate those feelings [of negative disconfirmation]."

Choice A is incorrect. This is the opposite of what the text claims. Rather, the researchers found that "many users" reversed "their initial sense of disappointment." Choice B is incorrect. This choice doesn't reflect the text. What the expectations of smart home tech users are is not discussed. Choice D is incorrect. This choice doesn't reflect the text. How often these topics have been studied is not mentioned.

Q5

D

- D. They show shapes consistent with monumental platforms and dense central settlements linked to smaller settlements by a system of canals and roadways.
-

Choice D is the best answer. This finding, if true, would support the archaeologists' conclusion. Dense central settlements linked to smaller ones would provide evidence of cities and suburbs—in other words, “a form of urbanism.”

Choice A is incorrect. This finding, if true, would weaken the archaeologists' conclusion. Widely separated, small settlements with jungle in between would support the long-held belief that the Casarabe culture “was characterized by a sparse, widely distributed population and little intervention in the surrounding wilderness.” Choice B is incorrect. This choice wouldn't support the researchers' conclusion. These large cities are located outside Casarabe territory, which doesn't show evidence of Casarabe urbanism. Choice C is incorrect. This finding wouldn't support the archaeologists' conclusion. Scattered small farms in jungle clearings are not good evidence to support the existence of cities (“a form of urbanism”).

Q6

A

- A. some disagreements between the model's simulations of Titan's precipitation and humidity and the moon's actual precipitation and humidity are to be expected.
-

Choice A is the best answer because it most logically completes the text's discussion of modeling the atmosphere of Titan. The text establishes that the methane mole fraction (the ratio of methane to other atmospheric components) influences precipitation, humidity, and other meteorological phenomena and that observational data for Titan are too lacking ("too sparse and discrepant") to fully determine the range of this measure across the moon's various atmospheric levels. The text indicates that because of this limitation, researchers developed an atmospheric model (IPSL) that uses a simplified approach: it applies a uniform methane mole fraction for the lowest atmospheric level. Since it is very unlikely that the actual methane distribution across atmospheric levels on Titan is uniform, and since the methane mole fraction directly affects precipitation and humidity, it logically follows that the model's simplified treatment of the methane mole fraction would lead to some disagreements between the model's simulations and Titan's actual precipitation and humidity.

Choice B is incorrect because the text makes it clear that the model *already* uses a single value for Titan's methane mole fraction instead of a range. Further, the text suggests that the single value is a simplification of what is likely a more complex reality, so there's no reason to assume that a single value would be more desirable than a range. Choice C is incorrect. The text strongly suggests that Titan's actual methane mole fraction differs from the uniform value used in the model, even though some of the model's outputs align closely with observations, but nothing in the text indicates how the actual value would differ. There's no reason to assume that the actual methane mole fraction is higher than what the model uses. The "lowest level" mentioned in the text refers to the atmosphere and not Titan's methane mole fraction. Choice D is incorrect because the text makes it clear that the model *doesn't* reflect the actual variations in the methane mole fraction across Titan's atmospheric levels, since observational data are too limited. Because the model uniformly applies a set methane mole fraction, no inconsistencies in the simulations would be attributable to actual variations in the moon's methane mole fraction.

Q7

B

B. Research fails to confirm a commonly held belief about dog breeds and behavior.

Choice B is the best answer because it most accurately states the main idea of the text. The text indicates that dog owners typically claim that some dog breeds are “more likely than others to have particular personality traits.” In other words, the text points out that a commonly held belief about dog breeds is that their personality traits are heritable. The text then states that Kathleen Morrill and colleagues undertook research about dog trait heritability and found that “behavior varies widely among dogs of the same breed.” Because Morrill and colleagues found evidence for variability rather than consistency in the behavior of dogs of the same breed, the statement that research fails to uphold a commonly held belief about dog breeds and behavior accurately reflects the main idea of the text.

Choice A is incorrect. Although the text mentions that humans have long intervened in dogs’ reproduction by intentionally crossbreeding certain dogs, it doesn’t argue that such intervention is essential to the existence of dog breeds. Choice C is incorrect because the text doesn’t discuss the popularity of any dog breeds; breeds are mentioned as having certain traits, but the text says nothing about the popularity of these breeds or traits. Choice D is incorrect. Although the text briefly mentions that Morrill and colleagues conducted a study about dog traits using both surveys and DNA sequencing, this is not the main focus of the text. The text concerns the study’s results about the heritability of dog traits, not the particular methodology used by Morrill and colleagues.

Q8

D

- D. ethylene was associated with a greater absolute change in ripening time at 14°C, 16°C, and 18°C than at 20°C.

Choice D is the best answer because it most effectively uses data from the graph to complete the student's conclusion about banana ripening time with and without ethylene at different temperatures. The graph shows that at 20°C, the gap between the two bars showing ripening times for ethylene-treated bananas and untreated bananas crosses fewer than 2 gridlines (from about 4 days for ethylene-treated bananas to about 5.5 days for untreated bananas). Meanwhile, the graph shows that at 14°C, 16°C, and 18°C, the gap between the bars crosses more than 2 gridlines (from about 8 days for treated bananas to about 11 days for untreated bananas at 14°C; from about 6 days for treated bananas to about 9.5 days for untreated bananas at 16°C; and from about 5.5 days for treated bananas to about 8.5 days for untreated bananas at 18°C). Since the gap between the bars at each of these temperatures crosses more than 2 gridlines, and since each of these gaps is larger than the gap between the bars at 20°C, it can be concluded that ethylene was associated with a greater absolute change in ripening time at 14°C, 16°C, and 18°C than at 20°C.

Choice A is incorrect because the graph shows that ethylene-treated bananas stored at 20°C ripen more quickly than ethylene-treated bananas stored at the other temperatures do (about 4 days at 20°C vs. about 5.5, 6, and 8 days at 18°C, 16°C, and 14°C, respectively) and that untreated bananas stored at 20°C ripen more quickly than untreated bananas stored at the other temperatures do (about 5.5 days at 20°C vs. about 8.5, 9.5, and 11 days at 18°C, 16°C, and 14°C, respectively). The information in the graph therefore indicates that storing bananas at 20°C speeds up ripening time relative to storing bananas at the other temperatures shown, not that this storage temperature slows ripening time. Choice B is incorrect because the graph shows that as temperature increases, the ripening time of untreated bananas decreases, from about 11 days at 14°C to about 5.5 days at 20°C, with no exceptions to this trend. The graph therefore shows that differences in temperature were associated with absolute differences in ripening time, not that there was no association between differences in temperature and differences in ripening time. Choice C is incorrect because the graph shows that ripening times of ethylene-treated bananas at 14°C and 16°C were about 8 and 6 days, respectively, which is greater than, not less than, ripening times of ethylene-treated bananas at 18°C and 20°C, which were about 5.5 and 4 days, respectively. In other words, bananas treated with ethylene ripen more slowly, not faster, at 14°C and 16°C than at 18°C and 20°C.

Q9

B

B. it does not allow for multiple castings of the same burrow over time.

Choice B is the best answer. Since resin casting “inevitably requires destroying the burrow,” it would be impossible to make multiple castings of the same burrow over time.

Choice A is incorrect. The passage doesn’t discuss size requirements for completing resin casting, so there’s no basis for this inference. Choice C is incorrect. The passage never mentions how long the casting process takes nor how long *A. bellulus* takes to construct a borrow, so there’s no basis for this inference. Choice D is incorrect. The passage never states that recovering the model distorts the resin’s shape, only that it destroys the burrow. Therefore, there’s no basis for this inference.

Q10

B

- B. Although theoreticians were justified in studying antigravity before the release of the 2023 report, the report's findings suggest that the rationale for theoretical consideration offered in the 2022 paper by Dimopoulos et al. is no longer applicable.

Choice B is the best answer because it most accurately states the main idea of the text. The text begins by acknowledging that in 2022, Dimopoulos et al. were justified in arguing for antigravity on theoretical grounds because although there was no experimental evidence supporting its existence at that time, there was no evidence contradicting its existence either. The text then introduces the 2023 report describing an experiment that produced observations "thoroughly inconsistent with antigravity." Thus, the main idea is that while the study of antigravity was theoretically justified before the 2023 report (due to lack of evidence against it), the report's findings undermine the rationale for such theoretical consideration since there is now evidence against this hypothetical phenomenon.

Choice A is incorrect. Although antihydrogen ballistics observations were conducted at CERN, the text provides no evidence that they were conducted to test conclusions in Dimopoulos et al.'s 2022 paper. Rather, the text presents the 2023 report as new evidence that contradicts the existence of the hypothetical phenomenon that Dimopoulos et al. had considered. Choice C is incorrect because the text doesn't suggest that Dimopoulos et al. assumed that unambiguous proof of antigravity would be unattainable. Rather, the text states that they argued antigravity was worth considering theoretically because evidence against it was lacking at that time. At the same time, the results in the 2023 report do not undermine but strengthen the idea that the proof of antigravity might not be achievable. Choice D is incorrect because the text doesn't suggest that there was an "inordinate focus on theoretical considerations" before 2023 or that such a focus hindered the development of experimental methods. In fact, the text describes Dimopoulos et al.'s position regarding the theoretical consideration of antigravity as "rightly argued," indicating approval of their approach given the absence of experimental evidence at that time.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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